Sari Pagurek van Mossel

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EDUCATION

- **Queen's University Bachelors of Computing Honours with Minor in Film and Media**
  - Achieved Degree with Distinction (Faculty of Arts and Sciences Dean's Honours List across all 4 years)
  - Selected to participate in credited exchange program at Uppsala Universitet, Uppsala, Sweden (January - June 2024)

PROFESSIONAL EXPERIENCE

Software Development Engineer L4, Amazon (Fulfillment Tech Robotics) — June 2025 - Current

- Led fullstack initiatives for large-scale scheduling systems that support high volume and production critical workflows, particularly using Java and React.js
- Built, deployed, and operated services using AWS technologies such as S3, EC2/ECS, IAM, and infrastructure-as-code CDK to enable reliable, scalable, near real-time data processing
- Drove a major platform migration with a successful phased rollout to the entire North American Sort Center population, spearheading cross-team communication and testing
- Diagnosed and resolved production issues as part of my team's on-call rotation, as well as implemented monitoring and alarms, and delivered targeted fixes to reduce high severity ticket volume and error spikes

Software Development Engineer Intern, Amazon (Fulfillment Tech Robotics) — June - Sept 2024

- Designed and implemented new full stack features for labour scheduling software utilizing React.js, Java with SpringMVC, and AWS tools including Kinesis and DynamoDB
- Delivered a complete end-to-end feature that enabled successful client migration and deprecation of a legacy service
- Applied software engineering best practices through automated unit, integration, and functional testing, code reviews, agile development, and technical documentation

Software Developer, Queen's Visual Cognition Lab (with Dr. Monica Castelhano) — May - Dec 2023

- Developed research and data analysis software for cognition studies alongside psychology graduate students and professors using Tobii VR Headset SDK
- Created and maintained Virtual Reality simulations using Unity and C# to collect and calculate fixation and saccadic movement data (precise eye movement metrics) for perception research
- Applied linear algebra concepts to transform 3D coordinate spaces and measure angles
- Designed and developed analysis software in Python using techniques including Hidden Markov Modelling, Switch-Point Analysis, and data processing with other Machine Learning libraries and open-source computer vision software
- Implemented analysis and calculation techniques from state-of-the-art research papers by collaborating professors

UX Design Intern, Goodself Co. — May 2022 - Aug 2023

- Held a lead responsibility in creating and updating app UI design features, prototypes, and user flows in Figma, specifically spearheading the design of 3 major app features
- Efficiently conducted quality assurance testing using git, bash shell scripting, and Xcode simulator, communicating effectively with both product and development teams

PROJECTS / RESEARCH

Computer Assisted Medial Patellofemoral Ligament (MPFL) Reconstruction: View on Github — 2025

- Collaborated with **Dr. James Stewart** to further his ongoing research in Computer Assisted MPFL Reconstruction surgery
- Computed a model of motion to simulate knee joint dynamics using principal curvatures and muscular forces, with the goal of minimizing potential complications such as graft impingement on surrounding anatomical structures and ensuring graft isometry
- Developed using C++ and OpenGL

3D to 2D: Using Image Segmentation to Automate Rotoscoping for Animation: View on Github — 2024

- Using a U-Net Convolutional Neural Network structure to process live footage frame-by-frame to simplify the subject into 3 discrete shades as well as separate it from the background in an effort to create a base rotoscoped animated sequence

Heatmap Display for Eye Tracking Data: View on Github — 2023

- Implemented OpenCV and other Python libraries to calculate frame by frame coordinates and generate a heatmap visualization from given eye movement fixation and saccade data

EXTRACURRICULAR / LEADERSHIP

- **Computing Department Teaching Assistant, Queen's University** — 2024-25
- **Research Team Member, QMind** — 2024-25
  - Collaborated on an undergraduate machine learning research paper to experiment with Gen AI and GANs (Generative Adversarial Networks) for super resolution photo reconstruction
- **Vice Chair of HackHer & Web Developer, Queen's University Women In Computing** — 2022-2025
- **Photographer, MUSE Magazine:** View Publication — 2023-24
- **Development Team Lead and UX Designer, Canadian Youth for Youth Empowerment:** View Work — 2023
- **Head of Photography, Vogue Charity Fashion Show:** View Work — 2022-23
- **Layout Designer, QUILT Undergraduate Literary Publication:** View Publication — 2022

SKILLS

Languages and Frameworks: HTML, CSS, Python, JavaScript, Java, Bash Shell Scripting, C, C#, C++, React.js, Processing, Git, OpenGL

Software and Tools: AWS Infrastructure (SQS, Kinesis, DynamoDB, EC2/ECS, S3, IAM, CDK), Figma, Adobe Creative Suite, Unity, Cinema4D, Unreal Engine, Xcode

Creative: 2D/3D Animation, Motion Graphics, Game Development

AWARDS

Creative Computing Showcase at Queen's University Best Art Project (2023)

HackHer (Queen's University Hackathon) First Place Category Winner in Food Insecurity & Social Good (2023)

Queen's University Principal's Scholarship for Academic Excellence (2021)

OCDSB Silver Medal given to averages of 90+ (2019-2021)

Ontario Scholar Award (2021)

Lisgar Collegiate Institute Michael Rust-Smith Memorial Award for Excellence in Arts and Science (2021)

Lisgar Collegiate Institute Award for Excellence in Visual Art (2021)